Scaling Relations and MRA

Course: Math 414

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Core Idea

At a high level, the process of processing (such as compressing or denoising) a signal $y = f(t)$ in practice is as follows:

1. **Sample**, we can use a power of two greater than Nyquist rate $2^J > 2v = \frac{\Omega}{\pi}$ for our purposes

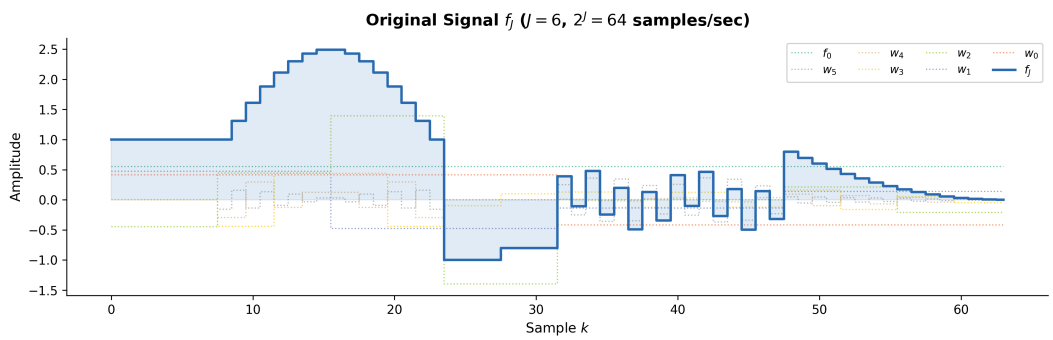

Figure 1: Sampling our function. Dotted lines are the decomposition as well

1. **Decompose** (with the algorithm below) f_J into a signal of the form $f_{J(x)} = \sum_{j=0}^{J-1} w_j + f_0$

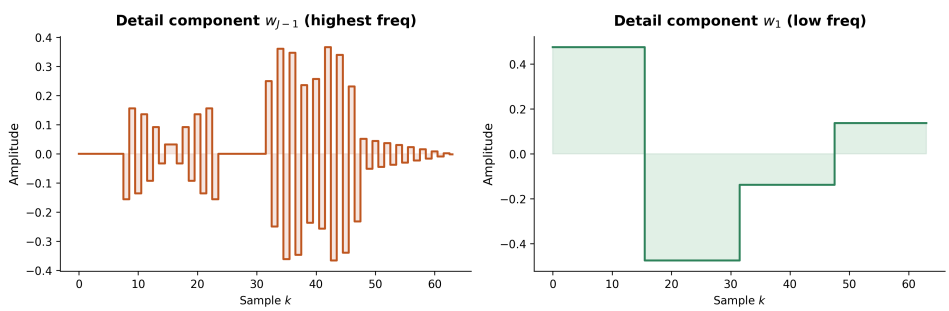

Figure 2: I just chose two random ones, the full decomposition is above

1. **Process** the signal.
   1. To filter out high frequencies, we can choose a max value of b_k^j. If we think about b_k as acting on ψ by increasing the height, then this means that those frequencies are gone
   2. To compress the signal, then b_k^j under a threshold (meaning they don't make a big difference), get thrown out

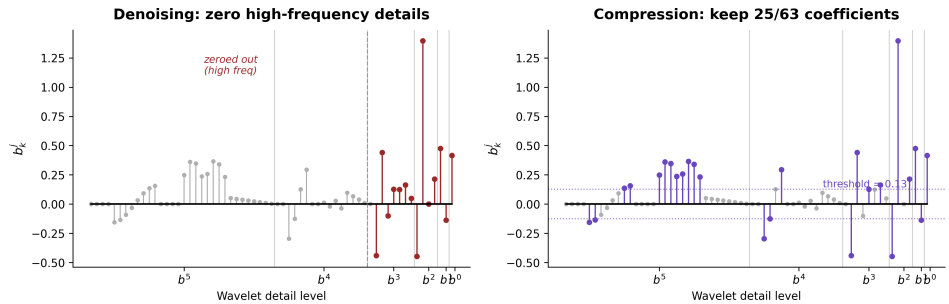

Figure 3: To remove high frequencies, we literally just remove the wavelets above a certain level. To compress, we remove wavelets which don't add much change

1. **Reconstruct** from our coefficients

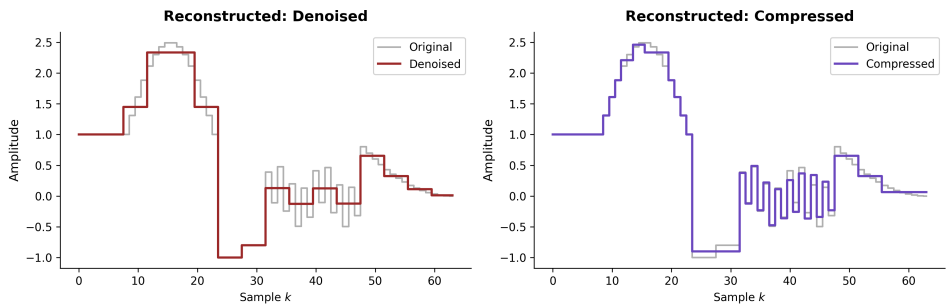

Figure 4: Just remaking the signal from our b_k that we got

Definitions

This notation is important

$$f_j \in V_j, f_j = \sum_{k \in \mathbb{Z}} a_k^j \underbrace{\phi(2^j x - k)}_{v_j} \tag{1}$$

Here is how we decompose our functions for the following proofs

$$\phi(x) = \phi(2x) + \phi(2x - 1) \tag{2}$$

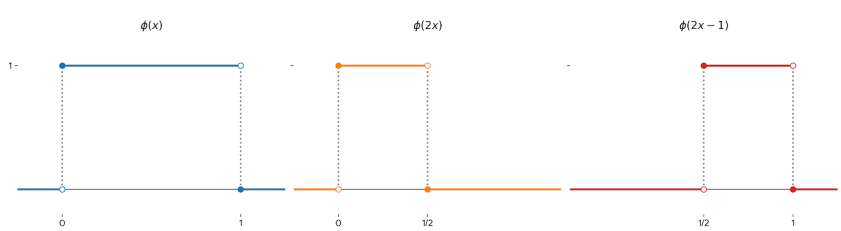

Figure 5: How we can express the haar scaling function

$$\psi(x) = \phi(2x) - \phi(2x - 1) \tag{3}$$

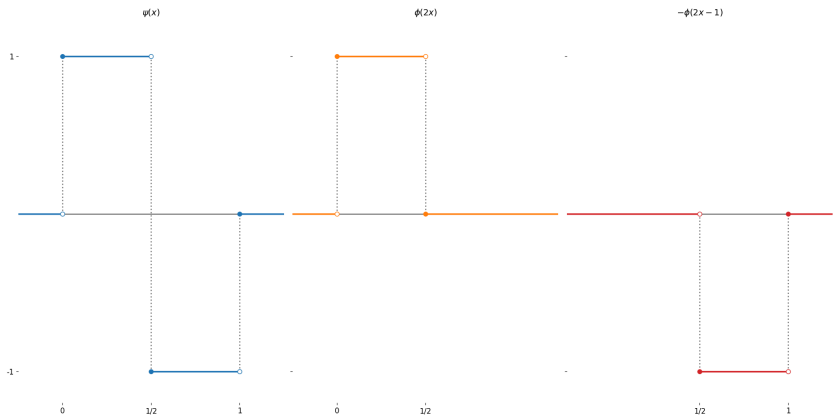

Figure 6: How we can express the haar wavelet function

If we add them

$$\phi(2x) = \frac{1}{2}(\phi(x) + \psi(x)) \tag{4}$$

If we subtract them

$$\phi(2x-1) = \frac{1}{2}(\phi(x) - \psi(x)) \tag{5}$$

If we replace x with $x-k$

$$\phi(x-k) = \phi(2x-2k) + \phi(2x-2k-1) \tag{6}$$

$$\psi(x-k) = \phi(2x-2k) - \phi(2x-2k-1) \tag{7}$$

$$\phi(2x-2k) = \frac{1}{2}(\phi(x-k) + \psi(x-k)) \tag{8}$$

$$\phi(2x-2k-1) = \frac{1}{2}(\phi(x-k) - \psi(x-k)) \tag{9}$$

If we replace x with $2^{j-1}x$ in Equation 4, Equation 5

$$\phi(2^j x) = \frac{1}{2}(\phi(2^{j-1}x) + \psi(2^{j-1}x)) \tag{10}$$

$$\phi(2^j x - 1) = \frac{1}{2}(\phi(2^{j-1}x) - \psi(2^{j-1}x)) \tag{11}$$

Haar Decomposition Formula

$$a_k^0 = \frac{1}{2}(a_{2k}^1 + a_{2k+1}^1) \tag{12}$$

$$b_k^0 = \frac{1}{2}(a_{2k}^1 - a_{2k+1}^1) \tag{13}$$

More generally

$$f_j = w_{j-1} + f_{j-1} \tag{14}$$

$$w_{j-1} = \sum_{k\in\mathbb{Z}} b_k^{j-1}\psi(2^{j-1}x - k) \in W_{j-1} \tag{15}$$

$$f_{j-1} = \sum_{k\in\mathbb{Z}} a_k^{j-1}\psi(2^{j-1}x - k) \in V_{j-1} \tag{16}$$

$$a_k^{j-1} = \frac{1}{2}\left(a_{2k}^j + a_{2k+1}^j\right) \tag{17}$$

$$b_k^{j-1} = \frac{1}{2}\left(a_{2k}^j - a_{2k+1}^j\right) \tag{18}$$

This gives us

$$f_j = w_{j-1} + w_{j-2} + ... + w_0 + f_0 \tag{19}$$

Haar Reconstruction Formula

$$(a_{2k}^1 = a_k^0 + b_k^0 \tag{20}$$

$$(a_{2k+1}^1 = a_k^0 - b_k^0 \tag{21}$$

$$f_J = \sum_{k\in Z} a_k^j \phi(2^J x - k) \tag{22}$$

Proof for decomposition

Let's suppose we have $f_1 \in V_1$, with O.N. basis $\{\phi(2x-k), k \in \mathbb{Z}\}$

$$f_1(x) = \sum_{k\in\mathbb{Z}} a_k^1 \phi(2x - k) \tag{23}$$

Split into even and odd

$$f_1(x) = \sum_{k\in\mathbb{Z}} a_{2k}^1 \phi(2x - 2k) + \sum_{k\in\mathbb{Z}} a_{2k+1}^1 \phi(2x - 2k - 1) \tag{24}$$

From Equation 8 and Equation 9, substitute them

$$f_1(x) = \sum_{k\in\mathbb{Z}} a^1_{2k}\left(\frac{1}{2}(\phi(x-k)+\psi(x-k))\right) + \sum_{k\in\mathbb{Z}} a^1_{2k+1}\left(\frac{1}{2}(\phi(x-k)-\psi(x-k))\right) \tag{25}$$

Collecting terms we have

$$f_1(x) = \sum_{k\in\mathbb{Z}}\left(\frac{1}{2}(a^1_{2k}+a^1_{2k+1})\phi(x-k) + \frac{1}{2}(a^1_{2k}-a^1_{2k+1})\psi(x-k)\right) \tag{26}$$

$$f_1(x) = \sum_{k\in\mathbb{Z}} \frac{1}{2}(a^1_{2k}+a^1_{2k+1})\phi(x-k) + \sum_{k\in\mathbb{Z}} \frac{1}{2}(a^1_{2k}-a^1_{2k+1})\psi(x-k) \tag{27}$$

But we also know from before that $f_1 = f_0 + w_0$

$$f_1(x) = \sum_{k\in\mathbb{Z}} a^0_k\phi(x-k) \sum_{k\in\mathbb{Z}} b^0_k\psi(x-k) \tag{28}$$

The coefficients from Equation 27 and Equation 28 are the same, and we get to the decomposition formula above.

Example

Suppose we have a signal $f_2 \in V_2$ represented by four scaling coefficients at level $j = 2$: $a^2 = (a^2_0, a^2_1, a^2_2, a^2_3) = [4, 2, 8, 6]$

This means our original sampled function can be written as: $f_2(x) = 4\phi(4x) + 2\phi(4x-1) + 8\phi(4x-2) + 6\phi(4x-3)$

1. Decomposition ($V_2 \to V_1 \to V_0$)
We use the Haar decomposition formulas $a^{j-1}_k = \frac{1}{2}\left(a^j_{2k} + a^j_{2k+1}\right)$ and $b^{j-1}_k = \frac{1}{2}\left(a^j_{2k} - a^j_{2k+1}\right)$.

Step 1: Down to V_1 $a^1_0 = \frac{1}{2}(a^2_0 + a^2_1) = \frac{1}{2}(4+2) = 3$

$$b^1_0 = \frac{1}{2}(a^2_0 - a^2_1) = \frac{1}{2}(4-2) = 1 \tag{29}$$

$a^1_1 = \frac{1}{2}(a^2_2 + a^2_3) = \frac{1}{2}(8+6) = 7$

$$b^1_1 = \frac{1}{2}(a^2_2 - a^2_3) = \frac{1}{2}(8-6) = 1 \tag{30}$$

At level V_1, our scaling coefficients are $a^1 = (3, 7)$ and our detail (wavelet) coefficients are $b^1 = (1, 1)$.

Step 2: Down to V_0 $a^0_0 = \frac{1}{2}(a^1_0 + a^1_1) = \frac{1}{2}(3+7) = 5$

$$b^0_0 = \frac{1}{2}(a^1_0 - a^1_1) = \frac{1}{2}(3-7) = -2 \tag{31}$$

Our fully decomposed signal is now represented by the coarsest average $a_0^0 = 5$ and the layered detail coefficients $b_0^0 = -2$ and $b^1 = (1, 1)$.

2. Reconstruction ($V_0 \to V_1 \to V_2$)

Now, let's reverse the process to retrieve our original signal using the Haar reconstruction formulas $a_{2k}^j = a_k^{j-1} + b_k^{j-1}$ and $a_{2k+1}^j = a_k^{j-1} - b_k^{j-1}$.

Step 1: Up to V_1 $a_0^1 = a_0^0 + b_0^0 = 5 + (-2) = 3$

$$a_1^1 = a_0^0 - b_0^0 = 5 - (-2) = 7 \tag{32}$$

We have perfectly recovered $a^1 = (3, 7)$ using our V_0 components.

Step 2: Up to V_2 $a_0^2 = a_0^1 + b_0^1 = 3 + 1 = 4$

$$a_1^2 = a_0^1 - b_0^1 = 3 - 1 = 2 \tag{33}$$

$a_2^2 = a_1^1 + b_1^1 = 7 + 1 = 8$

$$a_3^2 = a_1^1 - b_1^1 = 7 - 1 = 6 \tag{34}$$

We have successfully reconstructed our original signal coefficients $a^2 = (4, 2, 8, 6)$!